

EXPERIMENT 6

Student Name: Pranav Sood

UID: 24BDA70188

Branch: AIT-CSE

Section/Group: 24AIT_KRG-G2

Semester: 5

Subject Name: Computer Network

Subject Code: 24CSH-337

Experiment 6 – Implement VLAN and VLAN Trunking Protocols.

Aim: To implement Virtual Local Area Networks (VLANs) and VLAN trunking on network switches, and to configure communication between different VLANs using appropriate trunking protocols.

Objective

- To understand the concept and purpose of VLANs in a computer network.
- To create and configure multiple VLANs on a network switch.
- To assign switch ports to specific VLANs.
- To configure trunk links for carrying traffic from multiple VLANs.
- To understand the operation of VLAN trunking protocols such as IEEE 802.1Q.
- To verify connectivity and VLAN communication using suitable network commands.

Theory:

L2

Switch:

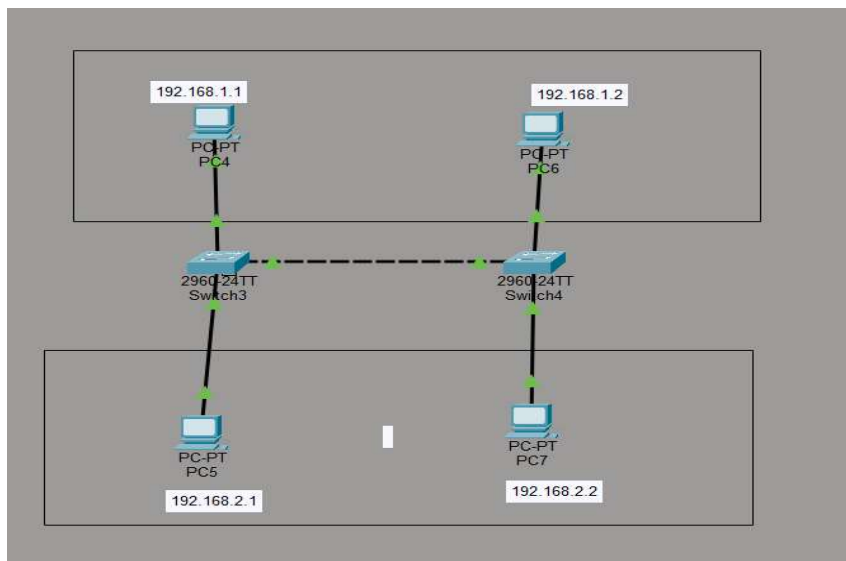
A Layer 2 switch operates at the Data Link Layer (Layer 2) of the OSI model and forwards Ethernet frames using MAC addresses. It learns the MAC addresses of connected devices and stores them in a MAC address table to determine the appropriate port for forwarding frames. L2 switches support VLANs, allowing a physical network to be divided into multiple logical broadcast domains. They can be configured with access ports for individual VLANs and trunk ports to carry traffic from multiple VLANs. However, an L2 switch cannot normally perform routing or communication between different VLANs.

EXPERIMENT 6

L3 Switch:

A Layer 3 switch operates at both the Data Link Layer (Layer 2) and Network Layer (Layer 3) of the OSI model. It performs normal switching using MAC addresses and can also route packets between different networks or VLANs using IP addresses and routing tables. L3 switches support Inter-VLAN Routing, usually through Switch Virtual Interfaces (SVIs), allowing devices belonging to different VLANs to communicate without requiring a separate router. They are commonly used in larger networks where both high-speed switching and routing capabilities are required.

Practical Steps:



Step 1: Take two switches 2960 Cisco and take 4 PC's.

Step 2: Use Copper-Straight Through cables to connect them.

PC1 → SW1 Fa0/1

PC2 → SW1 Fa0/2

PC3 → SW2 Fa0/1

PC4 → SW2 Fa0/2

Then connect two switches:

SW1 Gi0/1 → SW2 Gi0/1

EXPERIMENT 6

Step 3:

Assign IP Addresses:

PC1:

IP Address: 192.168.10.11

Subnet Mask: 255.255.255.0

PC3:

IP Address: 192.168.10.12

Subnet Mask: 255.255.255.0

PC1 and PC3 are going to belong to: VLAN 10

So their IP addresses are also from: 192.168.10.0

PC2:

IP Address: 192.168.20.11

Subnet Mask: 255.255.255.0

PC4:

IP Address: 192.168.20.12

Subnet Mask: 255.255.255.0

VLAN 20.

Step 4: Configuring VLAN on SW1 (Switch 1).

SW1 -> GO TO CLI.

Switch>

Enter Privileged Mode:

Type: Enable

Switch#

Enter Configuration Mode:

Type: Configure Terminal or Conf t

EXPERIMENT 6

Switch(config)#

Step 5: Create VLAN 10 on SW1

Type: VLAN 10

Switch(config-vlan)#

Give it a name: name STUDENTS

Now VLAN 10 has been created.

VLAN 10 -> STUDENTS.

Step 6: Create VLAN 20 on SW1

Type: Exit

Then: VLAN 20

And

Name FACULTY

Now SW1 has:

VLAN 10 -> STUDENTS

VLAN 20 -> FACULTY

Step 7: Assigning PORTS to VLAN.

Creating a VLAN does not automatically put PCs into that VLAN.

We still have to tell the switch: "Fa0/1 belongs to VLAN 10"

And "Fa0/1 belongs to VLAN 20"

Step 8: Configure Fa0/1 as an Access Port.

Type: Exit

Now enter the interface: interface fa0/1

You should see: Switch(config-if)#

Now tell the switch that this is an access port:

EXPERIMENT 6

Type: Switchport mode access

Then assign it to VLAN 10: switchport access vlan 10

So the complete configuration will be:

```
interface          fa0/1
switchport mode access
switchport access vlan 10
```

By this we tell SW1 that fa0/1 -> Access Port -> VLAN 10.

Therefore, PC1 -> Fa0/1 -> VLAN 10.

Step 9: Configure Fa0/2 for VLAN 20:

Type: Exit

```
interface fa0/2
then
switchport mode access
then
switchport access vlan 10.
```

Step 10: Repeat the same process for SW2.

Step 11: Now we have to configure the TRUNK.

This is the most important part of the experiment.

We have: SW1 Gi0/1 ===== Gi0/1 SW2

Currently, this is just a normal switch-to-switch connection.

We need to tell both switches:

"This link is going to carry multiple VLANs."

Therefore, we configure it as a trunk.

EXPERIMENT 6

Step 12:

On SW1:

enable

configure terminal

then

interface gigabitEthernet 0/1

then

switchport mode trunk

Now Gig0/1 is now the trunk port.

Step 13: Repeat the same process for SW2.

Learning Outcome

After completing this experiment, the learner will be able to:

1. Understand the basic concepts and working principles of Layer 2 and Layer 3 switches.
2. Configure and manage VLANs on network switches.
3. Configure trunk links to carry traffic between multiple VLANs.
4. Implement and verify **Inter-VLAN Routing** using a Layer 3 switch.